

Affordance Judgments in Mobile Augmented Reality with Cues

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ABSTRACT

We investigated two judgments of action capabilities with virtual objects presented through smartphones: passing through an aperture and stepping over a gap. The results showed that users were conservative in their affordance judgments for the two actions, but that judgments became more accurate with training by AR cues. In the post-cue trials, passing through judgments improved; in contrast, stepping over judgments became more precise when the cue was present, but did not display the same generalization in the post-cue block of improved estimates.

Index Terms: I.3.7 [Computer Graphics]: Three-Dimensional Graphics and Realism—Virtual Reality; J.4 [Computer Applications]: Social and Behavioral Sciences—Psychology

1 INTRODUCTION

AR applications for entertainment, education, healthcare, and training have been developed, and are now widely available in smartphones. One concern with the current technology is that virtual objects may not be perceived in the same way as real world objects. This difference, called a loss of *perceptual fidelity*, can lead to applications not working as well as they are intended to. This paper studies the perceptual fidelity of mobile AR by investigating how judgments of action possibilities with virtual objects are perceived. Possibilities for action in an environment are called *affordances*, proposed by Gibson [2]. Formally, they are observers' perceptions of their ability to act given the properties of the environment relative to their body's capabilities. Affordances can be used to provide a useful, objective measure of the perceptual fidelity of virtual or augmented environments.

Prior work on perceptual fidelity has been focused on immersive virtual environments shown in head-mounted displays (HMD), and most of this work did not evaluate observers' perception of their action capabilities [3,4]. In this paper, we test two action capabilities in mobile AR: passing through and stepping over, which have been well studied in the real world and in virtual reality. In our tasks, people make affordance judgments about passing through and stepping over virtual objects embedded in the real world as perceived through the smartphone. In addition, we allow for training of those judgments by showing an AR cue that indicates their true action capability. We test for whether the cue generalizes to future estimates by comparing the judgments prior to seeing the AR cue versus after experiencing the cue. We tested the following hypotheses across the two experiments:

- H1** The presence of an AR cue will make users' action judgments closer to their actual capabilities.
- H2** Training with an AR cue will improve users' action judgments in a post-cue block, compared to their baseline performance.

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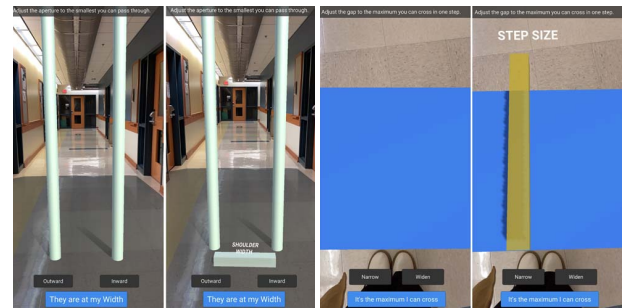
2 EXPERIMENTS

2.1 Materials

The applications for this experiment were coded in Unity (v.2019.2.21f). ARFoundation (v. 2.0.2) was used to support AR-Core and ARKit for enabling AR functionality on both the Android and iOS smartphone. There were two apps, one for each affordance, for each operating system published on the respective app repository. We deployed the apps in a distributed manner through the Apple App Store and Google Play Store. Thus, participants could be recruited widely, and run the apps in their local surroundings. An email account was coded in to send data files back as an attachment.

2.2 Participants

26 volunteers (12 females, 14 males) with a mean age of 32.57 years (range 19 to 68) completed Experiment 1. 26 volunteers (13 females, 13 males) with mean age of 32.61 years (range was 19 to 68) completed Experiment 2. All gave informed consent for the participation via an online consent form.



(a) Passing Through Experiment

(b) Stepping Over Experiment

Figure 1: Screenshot of passing through apertures (without the AR cue on left and with the AR cue on the right) and stepping over gap scenario.

2.3 Procedure

2.3.1 Experiment 1 : Passing Through Judgment

Once participants downloaded the app, they were given an overview of the experiment. They were then asked to measure the distance between the edge of each shoulder by following instructions and input the shoulder width (SW) in the app, as well as their age, gender and height. If the participant could not find a tape measure, the system set a default value for their SW to 0.5m (observed by Pointon et al. [5]). Before any judgments, they were required to find a clear open space, stand still and scan the ground with their phones to detect the plane thus generating a pair of virtual poles two meters in front of them.

Participants' task was to adjust the two poles by clicking on either the button "Inward" or "Outward" to be the width that they believed they could just pass through without turning their shoulders. First, participants completed a block of pre-cue trials which consisted of four judgments to assess baseline ability to make judgments (see Figure 1). The four adjustment trials included two "ascending" trials

(A; the poles started at 70% participants' SW) and two "descending" trials (D; 180% participants' SW), following the order: A, D, A, D. After completing the pre-cue block of trials, participants were asked to stay in their current location and were then presented with a virtual cue to help them better understand their body size (see Figure 1). This block also followed the pattern: A, D, A, D as in the pre-cue block. The only difference was the presence of an AR cue that represented the input SW. The cue was attached to the recognized ground plane and could be dragged by participants to help their judgments. To further assess whether the AR cue helped participants recalibrate their judgments of just-passable aperture width, a block of post-cue trials was conducted. Participants were instructed to take 4 steps forward and then to turn around and face the opposite direction. The walking and turning was implemented to limit participants' ability to use other cues in the environment to make their estimates, and also allowed us to assess whether training with the cue generalized to a new location. Again, this block followed the A, D, A, D pattern, but no AR cue was present.

After finishing all trials, participants were asked to rate the usefulness of the AR cue with a five-point Likert scale (ranging from extremely useless to extremely useful). Finally, participants clicked on the "Click to Finish" button to send data back to the experimenter.

2.3.2 Experiment 2 : Stepping Over Judgement

The procedure was similar to Experiment 1. First, participants were told to measure their maximum stride from trailing toe to leading heel. The system default setup was 0.88m (observed in [1]). There were also 12 trials over 3 blocks (pre-cue, cue, post-cue). The task was to adjust the gap to be the width that people believed they could step over with one step. The pre-cue block contained 4 trials to gauge baseline judgments of stepping over. Before starting, participants were asked to scan the ground plane to generate a virtual gap at their feet (see Figure 1). In the cue block, an AR cue that represented the extent of their step size was shown to help make the judgments (see Figure 1). As in Experiment 1, participants were asked to take 4 steps, turn around and face the opposite direction before the post-cue trials. When all trials were completed, participants rated the usefulness of the AR cue and then sent the data back.

3 RESULTS

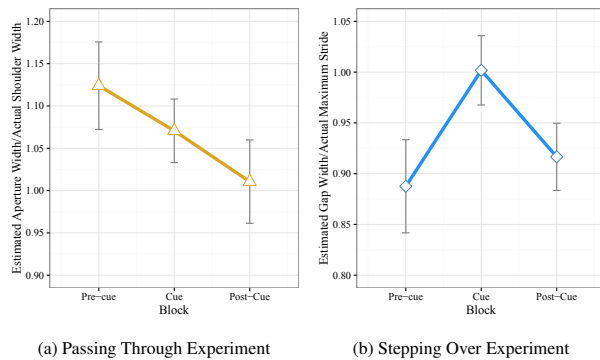


Figure 2: Average ratios for each block. Error bars show one standard error above and below the mean ratio.

For Experiment 1, we averaged the four trials from each block and calculated ratios of mean estimated just-passable width to measured SW. Means for each block with standard errors are shown in Figure 2. We ran a repeated-measures ANOVA on the ratios comparing the three blocks. There was a significant effect of block, $F(2,50) = 3.82, p < .03, \eta_p^2 = .13$. Planned contrasts comparing the cue and post-cue blocks to the pre-cue showed that ratios were not different

in the Cue ($M = 1.07$) compared to the pre-cue ($M = 1.12$) block, $F(1,25) = 1.78, p = .19$, but that ratios were lower in the Post-cue ($M = 1.01$) block compared to the pre-cue block, $F(1,25) = 8.11, p < .01, \eta_p^2 = .24$. These results suggest that providing the cue reduced estimates for passing through when the cue was removed in the post-cue trials. This finding supports the second hypothesis.

For the second experiment, one concern is whether the placement of the AR object is reliable relative to the detected location of the ground plane. We analyzed the mean altitude of the AR gap plane per participant in the pre-cue block versus the post-cue block, after the ground had been rescanned. A paired sample t-test on these means revealed no significant difference between these altitudes.

As in Experiment 1, we calculated ratios (see Figure 2) of estimated just-crossable gap to measured maximum stride. Three participants were excluded due to the inaccurate measurements. We ran a repeated-measures ANOVA on the ratios of the three blocks. There was a significant effect of block, $F(2,44) = 4.39, p < .02, \eta_p^2 = .17$. Planned contrasts comparing the cue and post-cue blocks to the pre-cue showed that estimations in the cue block ($M = 1.00$) were greater than in the pre-cue block ($M = .89$), $F(1,22) = 7.46, p < .012, \eta_p^2 = .25$. This finding supports our first hypothesis, that users estimates will become more accurate with the AR cue is present. However, there was no difference between the post-cue block ($M = .92$) and the pre-cue block, $F(1,22) = .54, p = .471$, which suggests that while the AR cue increased estimations of stepping capability during the cue block, the effect of the cue did not generalize when the cue was no longer present.

4 CONCLUSION

This paper provides a foundation for using applications with smartphones for testing users' perceived action capabilities with virtual objects in mobile AR. We clearly show that data collection with smartphones is feasible and reliable. We find that users can perceive their action capabilities, and in the case of passing through, they can also improve their affordance judgments given training with an AR cue. The lack of transfer of training with judgments of stepping over may have been due to a variety of factors related to the cameras in the phones, their field of view, or the means by which scanning is employed to make the judgment. Future work will need to assess how AR cues may improve perceptions of action capabilities with virtual targets across different tasks, but the initial findings presented here suggest the use of mobile AR with smartphones for assessing and training perception is possible.

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